

Transformation between elements and orbital state

Tuesday, September 22, 2026 10:39 AM

Lecture 7 cont'd

Computing elements from the state vector
Curtis section 4.4, algorithm 4.2

$$\begin{bmatrix} \vec{r} \\ \dot{\vec{r}} \end{bmatrix} \rightarrow \begin{bmatrix} a \\ e \\ i \\ \Omega \\ \omega \\ \theta(t) \end{bmatrix}$$

1. Angular momentum vector
2. Semilatus rectum
3. Eccentricity vector
4. Node direction

$$\vec{h} = \vec{r} \times \dot{\vec{r}}$$

$$p = h^2 / \mu$$

$$\vec{e} = \frac{\dot{\vec{r}} \times \vec{h}}{\mu} - \hat{r}$$

$$\hat{n} = \frac{\hat{k} \times \vec{h}}{|\hat{k} \times \vec{h}|}$$

$$e = |\vec{e}|$$

$$a = \left(\frac{2}{r} - \frac{v^2}{\mu} \right)^{-1} = \frac{p}{1-e^2}$$

Lecture 8 9/24/2026

RAAN and inclination

$$\Omega = \arctan(\hat{n} \cdot \hat{j}, \hat{n} \cdot \hat{i})$$

-or-

or -

$$\cos \Omega = n_I$$

$$\begin{aligned} \text{if } n_J < 0 &\Rightarrow \pi < \Omega < 2\pi \\ n_J > 0 &\Rightarrow 0 < \Omega < \pi \end{aligned}$$

Inclination

$$\cos i = \hat{h} \cdot \hat{k} \quad 0 \leq i \leq 180^\circ$$

Argument of perigee

Angle between the ascending node $\hat{n}$ and $\hat{e}$

$$\cos \omega = \hat{n} \cdot \hat{e}$$

$$\text{if } e_k < 0, \text{ then perigee below eq. plane}$$

$$\pi \leq \omega \leq 2\pi$$

True anomaly

$$\cos \theta = \hat{r} \cdot \hat{e} = \frac{1}{e} \left(\frac{p}{r} - 1 \right)$$

$$\text{If } \vec{r} \cdot \dot{\vec{r}} < 0 \text{ then -}$$

The true anomaly is below the apse line (major axis), so

$$\pi \leq \theta \leq 2\pi$$

$$\vec{r} \cdot \dot{\vec{r}} = r \dot{r}$$

These give us the orbital elements from the orbital state vector, position and velocity

Perifocal coordinates

Curtis section 2.10

Perifocal coordinates

Curtis section 2.10

- P points to perigee
- Q is perpendicular to P in the orbital plane
- W is in the direction of angular momentum

$$\hat{P} = \hat{e}$$

$$\hat{W} = \hat{h}$$

We will compute in the perifocal frame, then 3-1-3 rotation into ECI

Finding orbital state vector (position, velocity) from orbital elements

This can be thought of in two steps

1. Get position, velocity in perifocal coordinates
2. Transform from perifocal to ECI with 3-1-3 Euler angle rotation

$$\begin{aligned}\vec{r} &= r \cos \theta \hat{P} + r \sin \theta \hat{Q} \\ &= \frac{p \cos \theta}{1 + e \cos \theta} \hat{P} + \frac{p \sin \theta}{1 + e \cos \theta} \hat{Q}\end{aligned}$$

Velocity

$$\dot{\vec{r}} = (\dot{r} \cos \theta - r \dot{\theta} \sin \theta) \hat{P} + (\dot{r} \sin \theta + r \dot{\theta} \cos \theta) \hat{Q}$$

$$\begin{aligned}\dot{r} &= - \frac{p}{(1 + e \cos \theta)^2} (-e \sin \theta) \dot{\theta} \\ &= \dot{\theta} \frac{pe \sin \theta}{(1 + e \cos \theta)^2} = \dot{\theta} \frac{r^2}{p} e \sin \theta.\end{aligned}$$

$$h = r^2 \dot{\theta} \quad p = \frac{h^2}{\mu} \Rightarrow h = \sqrt{\mu p}$$

$$v_r = \dot{r} = \frac{h}{r^2} \frac{r^2}{p} e \sin \theta = \sqrt{\frac{\mu}{p}} e \sin \theta$$

$$v_\perp = r \dot{\theta} = \frac{h}{r} = \frac{h}{p} \frac{p}{r} = \sqrt{\frac{\mu}{p}} (1 + e \cos \theta)$$

$$v_t = r \dot{\theta} = \frac{h}{r} = \frac{h}{p} \frac{p}{r} = \sqrt{\frac{\mu}{p}} (1 + e \cos \theta)$$

$$\begin{aligned} \dot{\mathbf{r}} &= \left(\sqrt{\frac{\mu}{p}} e \sin \theta \cos \theta - \sqrt{\frac{\mu}{p}} (1 + e \cos \theta) \sin \theta \right) \hat{\mathbf{P}} \\ &\quad + \left(\sqrt{\frac{\mu}{p}} e \sin^2 \theta + \sqrt{\frac{\mu}{p}} (1 + e \cos \theta) \cos \theta \right) \hat{\mathbf{Q}} \\ &= \sqrt{\frac{\mu}{p}} \left[-\sin \theta \hat{\mathbf{P}} + (\cos \theta + e) \hat{\mathbf{Q}} \right] \end{aligned}$$

Motion in perifocal coordinates

$$\begin{aligned} \mathbf{r} &= \frac{p}{1 + e \cos \theta} (\cos \theta \hat{\mathbf{P}} + \sin \theta \hat{\mathbf{Q}}) \\ \dot{\mathbf{r}} &= \sqrt{\frac{\mu}{p}} \left[-\sin \theta \hat{\mathbf{P}} + (\cos \theta + e) \hat{\mathbf{Q}} \right] \end{aligned}$$

See Curtis equations (4.43), (4.44)

Rotate 3-1-3 into ECI

$$\mathcal{R}_z(-\Omega) \mathcal{R}_x(-i) \mathcal{R}_z(-\omega)$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(-i) & \sin(-i) \\ 0 & -\sin(-i) & \cos(-i) \end{bmatrix} \begin{bmatrix} \cos(-\omega) & \sin(-\omega) & 0 \\ -\sin(-\omega) & \cos(-\omega) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{r} = \mathcal{R}_z(-\Omega)\mathcal{R}_x(-i)\mathcal{R}_z(-\omega) \begin{bmatrix} \frac{p \cos \theta}{1 + e \cos \theta} \\ \frac{p \sin \theta}{1 + e \cos \theta} \\ 0 \end{bmatrix}$$

$$\dot{\mathbf{r}} = \mathcal{R}_z(-\Omega)\mathcal{R}_x(-i)\mathcal{R}_z(-\omega) \begin{bmatrix} -\sqrt{\frac{\mu}{p}} \sin \theta \\ \sqrt{\frac{\mu}{p}} (\cos \theta + e) \\ 0 \end{bmatrix}.$$

Curtis example 4.7

Singular conditions

Circular and equatorial orbits

Equinoctial elements

Classical equinoctial elements Broucke & Cefola in 1972

Elements

- Semimajor axis a
- Y component eccentricity vector: h
- X component eccentricity vector: k
- Y component of the inclination/node: p
- X component of the inclination/node: q
- Mean longitude: λ

$\dot{\lambda} = n$, mean motion

Equinoctial transformations
if classical elements are
defined

$$a = a$$

$$h = e \sin(\omega + \Omega)$$

$$k = e \cos(\omega + \Omega)$$

$$p = \tan\left(\frac{i}{2}\right) \sin \Omega$$

$$q = \tan\left(\frac{i}{2}\right) \cos \Omega$$

$$\lambda = M + \omega + \Omega$$

$$a = a$$

$$e = \sqrt{h^2 + k^2}$$

$$i = 2 \arctan\left(\sqrt{p^2 + q^2}\right)$$

$$\Omega = \text{atan2}(p, q)$$

$$\omega = \text{atan2}(h, k) - \Omega$$

$$M = \lambda - \omega - \Omega = \lambda - \text{atan2}(h, k)$$

Equinoctial Kepler equation

F = equinoctial eccentric longitude, the analogue of the eccentric anomaly

$$\lambda = F - h \cos F + k \sin F$$

Compute in-plane coordinates

$$\beta = \frac{1}{1 - \sqrt{1 - h^2 - k^2}}$$

Cartesian coordinates of position and velocity in the (unrotated) orbital plane

$$X_1 = a [(1 - h^2 \beta) \cos F + h k \beta \sin F - k]$$

$$Y_1 = a [(1 - k^2 \beta) \sin F + h k \beta \cos F - h]$$

Position components

$$\dot{X}_1 = \frac{\sqrt{\mu a}}{r} [h k \beta \cos F - (1 - h^2 \beta) \sin F]$$

$$\dot{Y}_1 = \frac{\sqrt{\mu a}}{r} [(1 - k^2 \beta) \cos F - h k \beta \sin F]$$

Velocity components

Rotate to inertial frame

$$\hat{f} = \frac{1}{1+p^2+q^2} \begin{bmatrix} 1-p^2+q^2 \\ 2pq \\ -2p \end{bmatrix}$$

$$\hat{g} = \frac{1}{1+p^2+q^2} \begin{bmatrix} 2pq \\ 1+p^2-q^2 \\ 2q \end{bmatrix}$$

$$\vec{r} = X_1 \hat{f} + Y_1 \hat{g}$$

$$\dot{\vec{r}} = \dot{X}_1 \hat{f} + \dot{Y}_1 \hat{g}$$

Transformation from orbital state to equinoctial elements

1. Compute angular momentum vector, eccentricity vector, semimajor axis
2. Equinoctial bases

$$\begin{bmatrix} \hat{h} \\ \hat{f} \\ \hat{g} \end{bmatrix} = \hat{h} \left. \begin{array}{l} \hat{f} \\ \hat{g} \end{array} \right\} \text{ in orbital plane}$$

$$f_x = 1 - \frac{w_x^2}{1+w_z}, \quad f_y = -\frac{w_x w_y}{1+w_z}, \quad f_z = -w_x$$

$$\hat{g} = \hat{h} \times \hat{f}$$

3. Calculate h, k, p, q

$$h = \vec{e} \cdot \hat{g}, \quad k = \vec{e} \cdot \hat{f}$$

$$p = \frac{w_x}{1+w_z}, \quad q = -\frac{w_y}{1+w_z}$$

4. Calculate the mean longitude

$$X_1 = \vec{r} \cdot \hat{f} \quad Y_1 = \vec{r} \cdot \hat{g}$$

7. Calculate the mean longitude

$$X_1 = \vec{r} \cdot \hat{f}, \quad Y_1 = \vec{r} \cdot \hat{g}$$

$$\cos F = k + \frac{(1 - k^2 \beta) X_1 - h k \beta Y_1}{a \sqrt{1 - h^2 - k^2}}$$

$$\sin F = \frac{(1 - h^2 \beta) Y_1 - h k \beta X_1}{a \sqrt{1 - h^2 - k^2}}$$

$$\lambda = F + h \cos F - k \sin F \quad \text{Mean longitude}$$

Equinoctial orbital elements to orbital state

- Compute the radial distance $r = a(1 - k \cos F - h \sin F)$
- $X_1 = a[(1 - h^2 \beta) \cos F + h k \beta \sin F - k]$
- $Y_1 = a[(1 - k^2 \beta) \sin F + h k \beta \cos F - h]$
- $\dot{X}_1 = \frac{na^2}{r} [h k \beta \cos F - (1 - h^2 \beta) \sin F]$
- $\dot{Y}_1 = \frac{na^2}{r} [(1 - k^2 \beta) \cos F - h k \beta \sin F]$

$$\vec{r} = X_1 \hat{f} + Y_1 \hat{g}$$

$$\dot{\vec{r}} = \dot{X}_1 \hat{f} + \dot{Y}_1 \hat{g}$$

Lecture 9
9/29/2026

Formulation	Primary Size Element	Position Angle	Primary Strength / Use Case
EqOE (Broucke & Cefola 1972)	Semimajor axis a	Mean Longitude (λ)	General theoretical development, analytical perturbation theories.
MEE (Walker et al 1985)	Semilatus rectum p	True Longitude (T)	Low-thrust trajectory optimization, steering law designs

Formulation	Primary Size Element	Position Angle	Primary Strength / Use Case
EqOE (Broucke & Cefola 1972)	Semimajor axis a	Mean Longitude (λ)	General theoretical development, analytical perturbation theories.
MEE (Walker et al. 1985)	Semilatus rectum p	True Longitude (L)	Low-thrust trajectory optimization, steering law designs.
AEqOE (Horwood et al. 2011)	Mean motion n	Mean Longitude (λ)	High-accuracy orbit determination and non-linear uncertainty propagation.
GEqOE (Baù et al. 2021)	ν (generalized n)	Generalized mean longitude	High-fidelity propagation under dominant J_2 oblateness.

- Modified equinoctial elements
Semilatus rectum instead of sma
True longitude

$$p = a(1 - e^2)$$

$$L = \Theta + \omega + \Omega$$

If these elements exist

- Alternate equinoctial elements
Use mean motion instead of sma
- Generalized equinoctial elements

$$n = \sqrt{\frac{\mu}{a^3}}$$

ν "generalized mean motion"

Based on the total energy (kinetic + potential energy with perturbation forces)

- Modified Rodriguez parameters - equinoctial elements

$$\sigma_1 = \tan\left(\frac{i}{4}\right) \sin \Omega$$

$$\sigma_2 = \tan\left(\frac{i}{4}\right) \cos \Omega$$